

# Performance of Hybrid Bonded-Bolted Joints between GFRP from Decommissioned Wind Turbine Blades and Duplex Stainless Steel for Marine Applications

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## ABSTRACT

This paper investigates hybrid bonded-bolted (HBB) joints between repurposed wind turbine blade glass-fiber-reinforced polymer (GFRP) and 2205 duplex stainless steel, addressing the critical need for reliable joining strategies to integrate decommissioned wind turbine blades into marine infrastructure. Experimental tests encompassing double-lap and single-lap configurations, various end-distance-to-diameter ( $e_1/d_0$ ) ratios, bolt arrangements and laminate layups were conducted. These were further supported by finite element analysis and comparative assessments with ASCE/SEI 74-23, CEN/TS 19101:2022 and T/CECS 692-2020. Results show that HBB joints follow a staged mechanism: the first peak load is mainly governed by adhesive-layer failure, whereas the bolted path provides post-adhesive mechanical redundancy and governs the final GFRP failure. For double-lap joints,  $e_1/d_0$  and bolt arrangement significantly influenced the interaction among adhesive edge stresses, bolt-hole contact and shear-out resistance. In single-lap blind-bolted joints, unidirectional  $0^\circ$  laminates primarily failed by rapid fiber-aligned shear-out, while multidirectional  $+45^\circ/0^\circ/-45^\circ$  laminates promoted more bearing-dominated deformation and larger opening gaps. Consequently, retaining blade inner-wall multidirectional laminates near the blind-bolt bearing zone can enhance the fail-safe behavior of the HBB joint. While existing design codes adequately predict the final shear-out mode, they fail to capture the adhesive-governed peak resistance. Therefore, this study proposes a staged assessment framework that explicitly separates adhesive-governed peak resistance from bolt-controlled residual capacity for designing future repurposed blade GFRP-DSS joints in marine reuse applications.

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## 1. Introduction

Surging computational demands in the AI era are further fueling the rapid expansion of global wind energy. The International Energy Agency (IEA) projects global installed capacity will exceed 8,000 GW under a 2050 net-zero scenario [1]. Concurrently, decommissioned wind turbine materials are projected to reach 43 million tonnes by 2050 [2]. This accelerating decommissioning requires urgent sustainable end-of-life strategies to supersede traditional landfilling and incineration. While waste management approaches are diversifying, this paper focuses on repurposing (structural reuse). This approach exploits their structural integrity and residual mechanical properties to fabricate lightweight infrastructural components, delivering superior environmental and economic benefits [3]. Recent studies highlight their broad application potential due to blades' lightweight and high-strength characteristics. In particular, their inherent design for harsh offshore environments offers substantial reuse potential in marine and energy engineering (e.g., small vessel hulls, wave energy converters, floating platforms), an area that remains largely conceptual (Fig.1).

Establishing reliable connections between the blade's glass fiber reinforced polymer (GFRP) and new primary steel structures remains the core technical challenge for effective load transfer and structural integrity. Consequently, dissimilar material joining, specifically composite-to-metal, is extensively researched, with current methodologies categorized into adhesive bonding, mechanical fastening, and hybrid joining. Adhesive bonding preserves structural integrity and lightweight characteristics, ensures uniform stress distribution without fiber disruption, and exhibits excellent stiffness and load transfer efficiency; however, its strength is highly environmentally sensitive [4,5]. Mechanical fastening offers superior manufacturing consistency, but drilling compromises fiber continuity and induces stress concentrations, resulting in lower

ultimate load capacity and stiffness [4,6,7]. Hybrid joining integrates both mechanisms, typically providing greater energy absorption than purely bonded joints and higher initial failure loads and tensile stiffness than purely bolted joints [8]. However, because the adhesive layer is significantly stiffer than the bolts, the latter remain largely unloaded until adhesive failure occurs; consequently, some studies report hybrid joints offer only a marginal 5% improvement over pure adhesive bonds [9]. Therefore, the efficacy of hybrid joints is context-dependent, necessitating systematic investigation into specific joint configurations, particularly for repurposed wind turbine blade applications.

To withstand primarily spanwise bending loads, the blade's spar cap utilizes high-fiber-volume-fraction unidirectional (UD)  $0^\circ$  laminates, supplemented by a minor proportion of multidirectional (MD)  $+45^\circ/0^\circ/-45^\circ$  laminates [10,11]. While exhibiting exceptional axial tensile properties, its significantly lower transverse and shear strengths render such composites highly susceptible to net-tension, shear-out, and longitudinal splitting failures in bolted joints [7]. Although projects like the Re-Wind Network have successfully implemented thick decommissioned GFRP sections in bridge girders and transmission poles, leveraging their high axial strength and thickness via engineering solutions like blind bolts to meet safety requirements [12,13], research on the failure mechanisms, damage evolution, and systematic design methodologies for various joint configurations in such highly oriented  $0^\circ$  UD GFRP remains scarce.

Driven by potential secondary applications, particularly in marine environments, this study systematically investigated joints between decommissioned GFRP and duplex stainless steel (DSS) across a four-level parametric framework: (1) joining method (focusing on hybrid joints, with pure adhesive and pure bolted configurations serving as comparative baselines); (2) joint geometry (double-lap and single-lap with blind bolts); (3) fastener quantity (single and double bolts); and (4) fiber orientation of GFRP (unidirectional  $0^\circ$  layups and multidirectional  $+45^\circ/0^\circ/-45^\circ$  layups). Furthermore, the GFRP laminates were modelled using three-dimensional continuum shell elements to account for the laminate architecture. Numerical simulations were conducted on the double-lap hybrid joints to elucidate the GFRP failure mechanism, and on the single-lap joints to compare the effects of unidirectional  $0^\circ$  and multidirectional fibre layups on the failure mode. The experimental results are compared against existing composite joint capacity formulations to establish design criteria suitable for structural repurposing.



Fig. 1 Major application scenarios of repurposing strategy: (a): [12], (b): [14], (c): [13], (d): [11], (e): [3], (f): [15], (g) and (h): [16], (i) and (j): [17].

## 2. Experimental program

### 2.1. Material preparation and specimen manufacturing

A decommissioned 54-m, 11,875-kg DT112-2000/A7 wind turbine blade (Nantong Dongtai New Energy Equipment Co., China), originally from a 2MW-XE112-2000 turbine (XEMC Windpower Co., China), was utilized. Manufactured via vacuum infusion, the blade consists primarily of E-glass fiber-reinforced epoxy composites. Prior to decommissioning, it operated for approximately 10 years at a wind farm in Yongzhou, Hunan, China (elevation 260.5 m). Despite prolonged exposure to harsh environments, including severe thunderstorms, retirement resulted from systemic technological upgrades rather than structural failure. The metal components comprised 2205 duplex stainless steel (UNS S32205) with 612 MPa yield and 850 MPa tensile strengths. Fastening utilized M16 grade 316 stainless steel standard and expansion blind bolts. Based on comparative properties [4] and adherend compatibility, Araldite® 2015-1, a toughened structural epoxy was selected. Lap geometry and adhesive thickness were optimized via theoretical calculations and literature references, while fabrication and testing were maintained at a consistent 27 °C ambient temperature.

Joint specimens were categorized by geometric configuration (double-lap or single-lap with blind bolts), joining method (adhesive-only, only-bolted, or hybrid bonded-bolted), bolt arrangement (horizontal or vertical) and fiber orientation. Each group contained two to three replicates. Dimensional details, fabrication processes, and comprehensive experimental groupings are detailed in Figs. 2-4 and Table 1. Crucially, during the fabrication of all bonded specimens, 0.5 mm diameter steel beads were embedded within the uniformly applied adhesive to guarantee a consistent bondline thickness. Furthermore, because adhesive spew fillets intrinsically enhance overall joint strength [18], they were intentionally retained, circumventing the need for supplementary removal procedures.

### 2.2. Test setup

Following ASTM D5868 and ASTM D5961/D5961M standards [19,20], quasi-static uniaxial tensile tests were performed at 27 °C and 72% relative humidity. All joint configurations were tested on a 2000 kN capacity machine equipped with wedge grips at a constant 2 mm/min loading rate to extract load-displacement and load response (Fig. 5). Relative displacement between the GFRP and DSS plates was measured via a laser displacement sensor (Model BLG-30N-485, BOJKE).

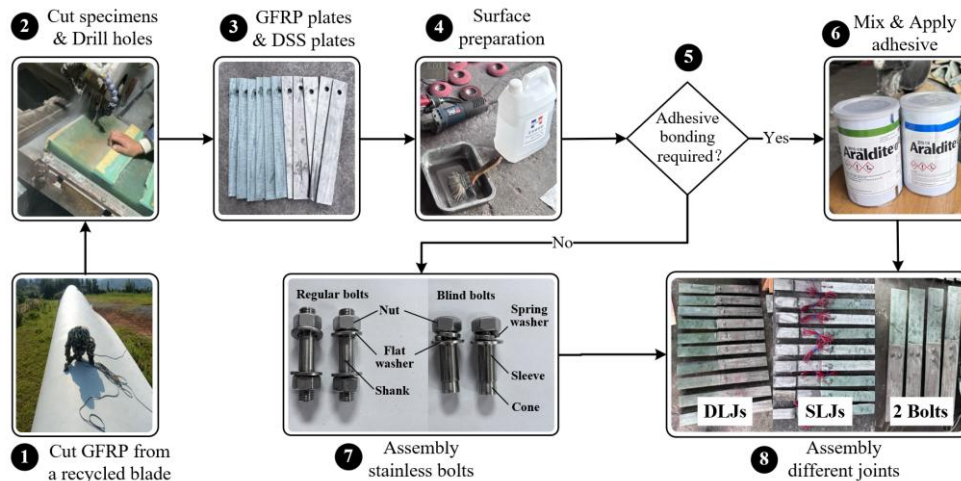


Fig. 2 Specimen manufacturing flowchart.

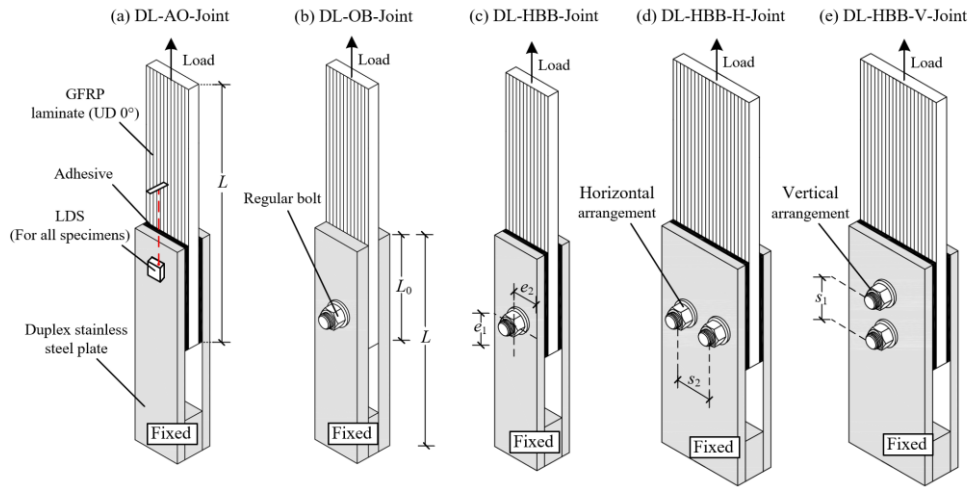


Fig. 3 Schematic diagrams of double-lap joints specimen.

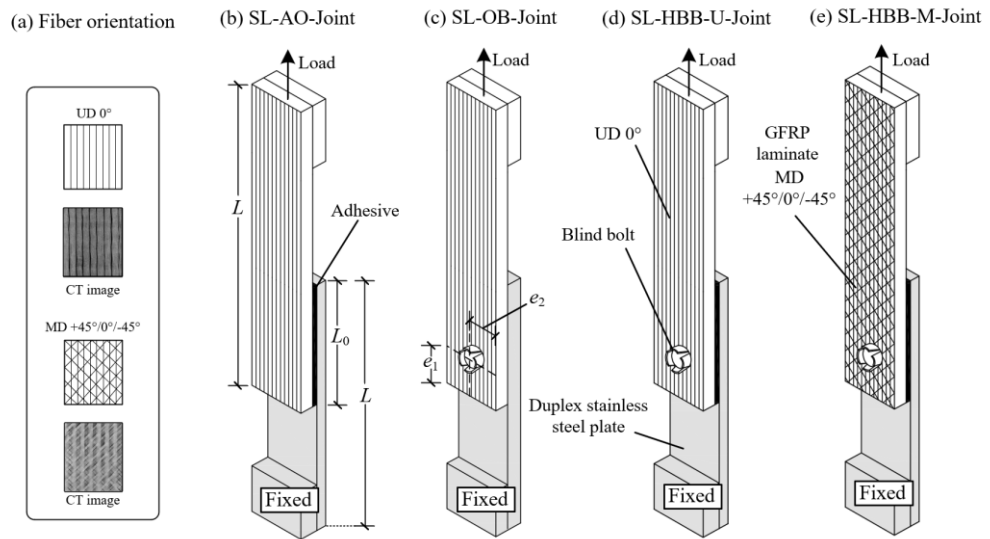


Fig. 4 Schematic diagrams of single-lap joints specimen.

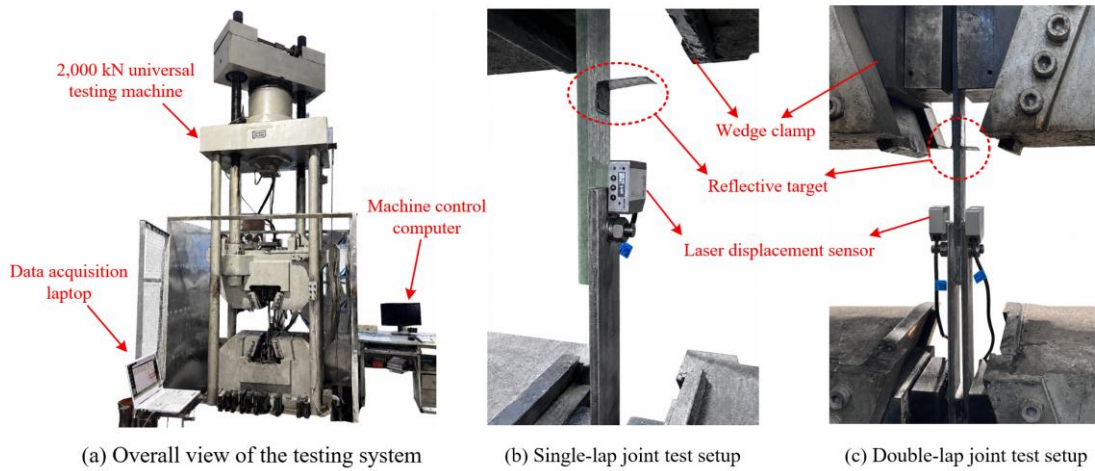


Fig. 5 Experimental test setup

Table 1 Geometrical dimensions for joints specimens.

| Specimen     | Dimensions               |                           |       |       |      |      |       |                    |
|--------------|--------------------------|---------------------------|-------|-------|------|------|-------|--------------------|
|              | $e_1/d_0$                | $e_2/d_0$                 | $d_0$ | $t_a$ | $t$  | $L$  | $L_0$ | $A_{net}$          |
|              |                          |                           | (mm)  | (mm)  | (mm) | (mm) | (mm)  | (mm <sup>2</sup> ) |
| DL-AOJ-1     | 0.0                      | 0.0                       | 16    | 0.5   | 10   | 400  | 88    | 10,560             |
| DL-AOJ-2     | 0.0                      | 0.0                       |       |       |      | 400  | 88    | 10,560             |
| SL-AOJ-1     | 0.0                      | 0.0                       |       |       |      | 400  | 88    | 5,280              |
| SL-AOJ-2     | 0.0                      | 0.0                       |       |       |      | 400  | 88    | 5,280              |
| SL-AOJ-3     | 0.0                      | 0.0                       |       |       |      | 400  | 88    | 5,280              |
| DL-OBJ-30-1  | 3.0                      | 1.88                      | 16    | 0.0   | 10   | 400  | 88    | 10,359             |
| DL-OBJ-30-2  | 3.0                      | 1.88                      |       |       |      | 400  | 88    | 10,359             |
| SL-OBJ-30-1  | 3.0                      | 1.88                      |       |       |      | 400  | 88    | 5,180              |
| SL-OBJ-30-2  | 3.0                      | 1.88                      |       |       |      | 400  | 88    | 5,180              |
| SL-OBJ-30-3  | 3.0                      | 1.88                      |       |       |      | 400  | 88    | 5,180              |
| DL-HBB-30-1  | 3.0                      | 1.88                      | 16    | 0.5   | 10   | 400  | 88    | 10,359             |
| DL-HBB-30-2  | 3.0                      | 1.88                      |       |       |      | 400  | 88    | 10,359             |
| DL-HBB-24-1  | 2.4                      | 1.88                      |       |       |      | 400  | 88    | 10,359             |
| DL-HBB-24-2  | 2.4                      | 1.88                      |       |       |      | 400  | 88    | 10,359             |
| DL-HBB-15-1  | 1.5                      | 1.88                      |       |       |      | 400  | 88    | 10,359             |
| DL-HBB-15-2  | 1.5                      | 1.88                      |       |       |      | 400  | 88    | 10,359             |
| SL-HBB-30-U1 | 3.0                      | 1.88                      |       |       |      | 400  | 88    | 5,180              |
| SL-HBB-30-U2 | 3.0                      | 1.88                      |       |       |      | 400  | 88    | 5,180              |
| SL-HBB-30-U3 | 3.0                      | 1.88                      |       |       |      | 400  | 88    | 5,180              |
| SL-HBB-30-M1 | 3.0                      | 1.88                      |       |       |      | 400  | 88    | 5,180              |
| SL-HBB-30-M2 | 3.0                      | 1.88                      |       |       |      | 400  | 88    | 5,180              |
| DL-HBB-30-V1 | 3.0 ( $s_1/d_0 = 3.38$ ) | 1.88                      |       |       |      | 450  | 140   | 33,198             |
| DL-HBB-30-V2 | 3.0 ( $s_1/d_0 = 3.38$ ) | 1.88                      |       |       |      | 450  | 140   | 33,198             |
| DL-HBB-55-H1 | 5.5                      | 3.75 ( $s_2/d_0 = 3.38$ ) |       |       |      | 450  | 140   | 33,198             |
| DL-HBB-55-H2 | 5.5                      | 3.75 ( $s_2/d_0 = 3.38$ ) |       |       |      | 450  | 140   | 33,198             |

Note: DL: double-lap; SL: single-lap; AOJ: adhesive-only joint; OBJ: only-bolted joint; HBB: hybrid bonded-bolted joint; U: unidirectional 0°; M: multidirectional +45°/0°/−45°; V: vertical bolt arrangement; H: horizontal bolt arrangement.

### 3. Finite element modeling

#### 3.1. Geometry and meshing

To elucidate the failure mechanisms, numerical models of the hybrid joints were developed in Abaqus/Explicit. The GFRP laminate was modelled using three-dimensional continuum shell elements (SC8R) to represent the laminate architecture and fibre orientations, while the DSS plates and bolts were discretized using solid elements (C3D8R). The adhesive layer was modelled with 3D cohesive elements (COH3D8) governed by a traction–separation law. Following a mesh convergence study, a non-uniform meshing strategy was adopted: the critical load-transfer regions, including the bolt-hole periphery and overlap edges, were locally refined to a minimum element size of 1 mm, whereas the far-field regions were meshed with a maximum element size of 10 mm. Geometric symmetry was exploited where applicable, and the corresponding symmetry boundary conditions were imposed.

#### 3.2. Material constitutive models

The intralaminar response of the GFRP was modeled using a three-dimensional strain-based Hashin criterion continuum

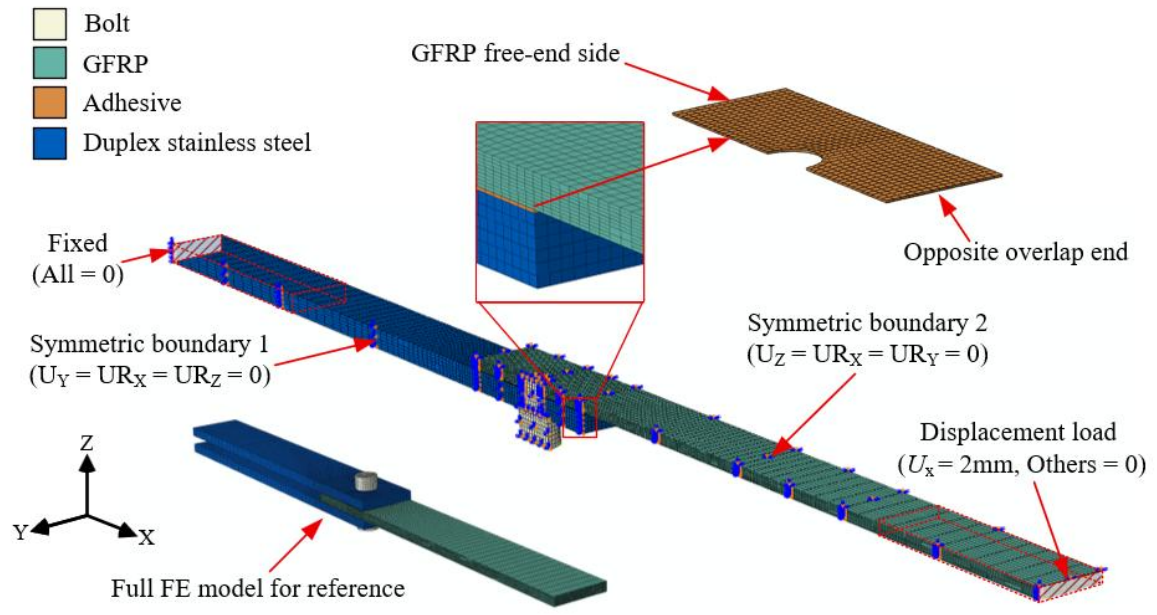
damage mechanics (CDM) formulation implemented in Abaqus/Explicit through a VUMAT subroutine, following the general strategy of user-defined progressive damage modeling for composite structures [21]. Based on the Hashin criterion [22], the fiber-dominated and matrix-dominated damage initiation modes were evaluated separately. In the present VUMAT, six damage-initiation indices were considered, including fiber tension, fiber compression, matrix tension, matrix compression, in-plane shear, and out-of-plane shear. To mitigate the influence of contact-induced local stress oscillations around the bolt hole, the initiation indices were formulated using strain-based measures rather than stress-based quantities. All damage-initiation thresholds were defined using the nominal GFRP material failure parameters listed in Table 3. After damage initiation, irreversible damage variables were progressively updated and introduced into a product-form stiffness degradation law to describe the post-initiation load transfer and damage growth [23]. Element deletion was not triggered immediately by the Hashin-type initiation flags, but was delayed until severe accumulated damage was reached. The adhesive layer was modeled using a bilinear traction–separation law governed by a quadratic nominal stress initiation criterion and energy-based damage evolution.

### 3.3. Interactions, boundary conditions and analysis settings

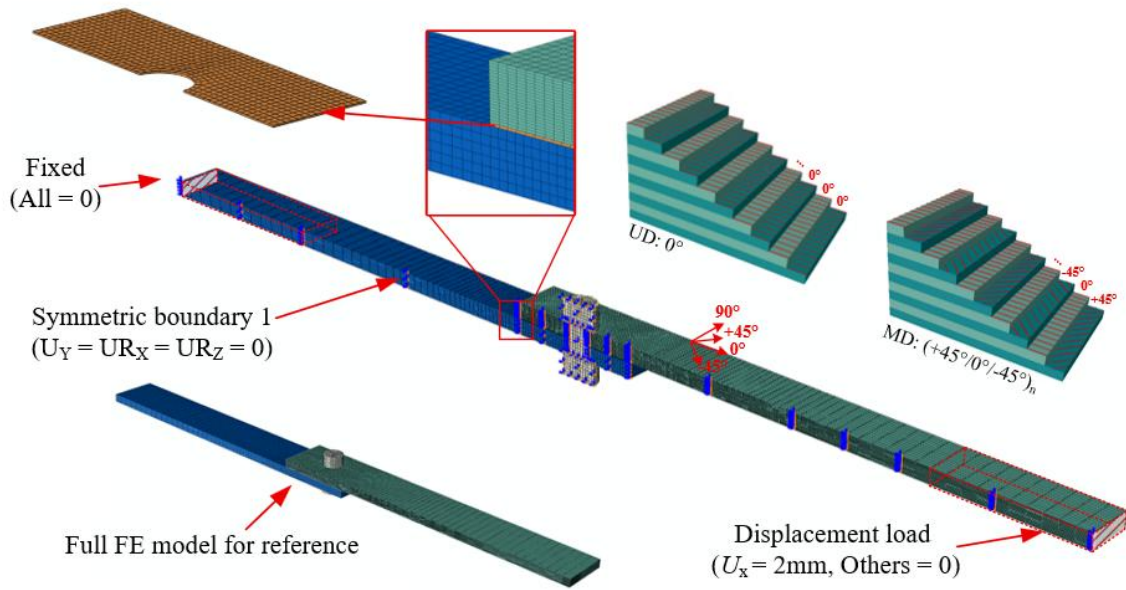
All contact interfaces employed a general contact algorithm, and normal behavior was defined as hard contact, while tangential interactions utilized a penalty friction formulation with a coefficient of  $\mu = 0.3$ . Quasi-static simulations proceeded in two consecutive steps. First, bolt preload was induced via a calibrated, predefined temperature drop applied to the bolt shank. Second, replicating experimental boundary conditions (120 mm gripped regions at both extremities), a displacement-controlled tensile load was applied to one end while the other remained fully constrained. Total reaction forces were extracted from the loading-end reference point. To ensure strict comparability with empirical load-displacement responses, numerical relative displacements were derived mirroring the experimental laser sensor configuration by calculating the displacement differential between the DSS terminus and a GFRP measurement point located 120 mm away (Fig. 6).

Table 2 Strain-based Hashin-type damage initiation and progressive degradation model for GFRP.

| Damage Mode                                                           | Damage initiation index                                                                                                                                                                                                                          | Progressive degradation                                                                                                                                                                                                                                                      |
|-----------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Fiber tensile failure<br>( $\epsilon_{11} > 0$ )                      | $F_{ft} = \left(\frac{\epsilon_{11}}{X_T^e}\right)^2 + k \left(\frac{\epsilon_{12}}{S_{12}^e}\right)^2 + k \left(\frac{\epsilon_{13}}{S_{13}^e}\right)^2 \geq 1$                                                                                 | $d_{ft} = \min\left(d_{max,t}, \max\left(d_{ft}^{old}, d_{max,t} \frac{\sqrt{F_{ft}} - 1}{r_{max,ft} - 1}\right)\right)$<br>$E_{11}^d = E_{11}^0 (1 - d_{ft})(1 - d_{fc})$                                                                                                   |
| Fiber compressive failure<br>( $\epsilon_{11} < 0$ )                  | $F_{fc} = \left(\frac{\epsilon_{11}}{X_C^e}\right)^2 \geq 1$                                                                                                                                                                                     | $d_{fc} = \min\left(d_{max,c}, \max\left(d_{fc}^{old}, d_{max,c} \frac{\sqrt{F_{fc}} - 1}{r_{max,fc} - 1}\right)\right)$                                                                                                                                                     |
| Matrix tensile failure<br>( $\epsilon_{22} + \epsilon_{33} \geq 0$ )  | $F_{mt} = \left(\frac{\epsilon_{22} + \epsilon_{33}}{Y_T^e}\right)^2 + \frac{\epsilon_{23}^2 - \epsilon_{22}\epsilon_{33}}{(S_{23}^e)^2} + \left(\frac{\epsilon_{12}}{S_{12}^e}\right)^2 + \left(\frac{\epsilon_{13}}{S_{13}^e}\right)^2 \geq 1$ | $d_{mt} = \min\left(d_{max,t}, \max\left(d_{mt}^{old}, d_{max,t} \frac{\sqrt{F_{mt}} - 1}{r_{max,mt} - 1}\right)\right)$<br>$E_{22}^d = E_{22}^0 (1 - d_{ft})(1 - d_{fc})(1 - d_{mt})(1 - d_{mc})$<br>$E_{33}^d = E_{33}^0 (1 - d_{ft})(1 - d_{fc})(1 - d_{mt})(1 - d_{mc})$ |
| Matrix compressive failure<br>( $\epsilon_{22} + \epsilon_{33} < 0$ ) | $F_{mc} = \left(\frac{\epsilon_{22} + \epsilon_{33}}{Y_C^e}\right)^2 + \left(\frac{\epsilon_{12}}{S_{12}^e}\right)^2 + \left(\frac{\epsilon_{13}}{S_{13}^e}\right)^2 + \left(\frac{\epsilon_{23}}{S_{23}^e}\right)^2 \geq 1$                     | $d_{mc} = \min\left(d_{max,c}, \max\left(d_{mc}^{old}, d_{max,c} \frac{\sqrt{F_{mc}} - 1}{r_{max,mc} - 1}\right)\right)$                                                                                                                                                     |
| In-plane shear failure                                                | $F_{s,12} = \left(\frac{\epsilon_{12}}{S_{12}^e}\right)^2 \geq 1$                                                                                                                                                                                | $d_{s,12} = \min\left(d_{max,c}, \max\left(d_{s,12}^{old}, d_{max,c} \frac{\sqrt{F_{s,12}} - 1}{r_{max,s12} - 1}\right)\right)$<br>$G_{12}^d = G_{12}^0 \Phi_{shear}(1 - d_{s,12})$                                                                                          |
| Out-of-plane shear failure                                            | $F_{s,23} = \left(\frac{\epsilon_{23}}{S_{23}^e}\right)^2 + \left(\frac{\epsilon_{13}}{S_{13}^e}\right)^2 \geq 1$                                                                                                                                | $d_{s,23} = \min\left(d_{max,c}, \max\left(d_{s,23}^{old}, d_{max,c} \frac{\sqrt{F_{s,23}} - 1}{r_{max,s23} - 1}\right)\right)$<br>$G_{13}^d = G_{13}^0 \Phi_{shear}(1 - d_{s,23})$<br>$G_{23}^d = G_{23}^0 \Phi_{shear}(1 - d_{s,23})$                                      |



(a) Model of double-lap HBB joint.



(b) Model of single-lap HBB joint.

Fig. 6 Symmetric FE Model details of the double-lap HBB joint.

Table 3 Properties of GFRP from a decommissioned blade and Adhesive (Araldite® 2015).

| Material       | Property                                | Symbol                            | Value                 | Unit                |
|----------------|-----------------------------------------|-----------------------------------|-----------------------|---------------------|
| GFRP           | Density                                 | $\rho_{\text{GFRP}}$              | $1.92 \times 10^{-9}$ | Ton/mm <sup>3</sup> |
|                | Elastic modulus                         | $E_{11}$                          | 42,700                | N/mm <sup>2</sup>   |
|                |                                         | $E_{22} = E_{33}$                 | 12,072                | N/mm <sup>2</sup>   |
|                | Poisson's ratio                         | $\nu_{23}$                        | 0.2100                | -                   |
|                |                                         | $\nu_{12} = \nu_{13}$             | 0.2629                | -                   |
|                | Longitudinal Tensile Failure Strain     | $\epsilon_1^T$                    | 0.0257                | -                   |
|                | Longitudinal Compressive Failure Strain | $\epsilon_1^C$                    | 0.0741                | -                   |
|                | Transverse Tensile Failure Strain       | $\epsilon_2^T$                    | 0.0059                | -                   |
|                | Transverse Compressive Failure Strain   | $\epsilon_2^C$                    | 0.0264                | -                   |
| Araldite® 2015 | In-Plane Shear Failure Strain           | $\gamma_{12}$                     | 0.0485                | -                   |
|                | Out-of-Plane Shear Failure Strain       | $\gamma_{13} = \gamma_{23}$       | 0.0222                | -                   |
|                | Resin system                            | -                                 | Expoxy                | -                   |
|                | Power Law exponent                      | -                                 | 1                     | -                   |
|                | Poisson's ratio                         | $\nu$                             | 0.33                  | -                   |
|                | Young's modulus                         | $E$                               | 1850                  | N/mm <sup>2</sup>   |
|                | Shear Modulus                           | $G$                               | 560                   | N/mm <sup>2</sup>   |
|                | Tensile failure strength                | $\sigma_y$                        | 21.63                 | N/mm <sup>2</sup>   |
|                | Shear failure strength                  | $\tau_y$                          | 17.9                  | N/mm <sup>2</sup>   |
|                | Normal Mode Fracture Energy             | $G_{\text{IC}}$                   | 0.43                  | N/mm                |
|                | Shear Mode Fracture Energy              | $G_{\text{IIC}}, G_{\text{IIIC}}$ | 4.7                   | N/mm                |

## 4. Results and discussion

### 4.1. Double-lap joints

#### 4.1.1. Load-displacement responses and failure modes

Figs 7 and 8 present the failure modes of single-joining-method joints and hybrid bonded-bolted joints. Initially, double-lap adhesive-only joints predominantly exhibited cohesive failure, leaving partial white residue at the interface alongside localized GFRP fiber tear (Fig. 7, DL-AOJ). Conversely, the ultimate failure mode for double-lap bolted joints was shear-out (Fig. 7, DL-OBJ). Fig 9 elucidates the progressive failure of GFRP in DL-OBJ joints: the initial peak signifies bearing failure, characterized by localized crushing at the hole-bolt contact; the subsequent peak corresponds to shear-out failure, generating two distinct shear planes (Fig. 7, DL-OBJ) as the bolt is continuously extruded. Consequently, evaluating both single-method joints, adhesive bonding provides higher stiffness but pronounced brittleness. For only- bolted joints, despite multi-stage failure characteristics, the ultimate GFRP shear-out is fundamentally brittle; structural integrity was entirely compromised following initial bearing failure. Therefore, further investigation into hybrid joints is warranted to evaluate their combined performance.

In double-lap hybrid bonded-bolted joints, failure modes consistently comprised cohesive failure in adhesive and shear-out in GFRP (Fig. 8). The joint initially reaches a peak load; a comparison with the adhesive-only joint curves indicates that this peak is primarily governed by the adhesive layer. Subsequently, the load drops rapidly and plateaus between 20 and 30 kN as the shear-out process occurs. This phase corresponds directly to the shear-out segment of the bolted-only joint curves. Specimen DL-HBB-30, featuring a larger end-distance-to-diameter ( $e_1/d_0$ ) ratio, exhibited a comparatively smaller relative displacement and ultimate load at failure. Therefore, the specific effect of the end distance requires further discussion.

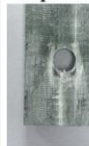
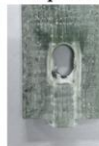
| Adhesive-only joints                                                              |                                                                                   |                                                                                   |                                                                                   |                                                                                   |                                                                                    |                                                                                    |                                                                                     |                                                                                     |                                                                                     |
|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| Adherend failure                                                                  |                                                                                   |                                                                                   |                                                                                   |                                                                                   |                                                                                    | Cohesive & Mixed failure                                                           |                                                                                     |                                                                                     |                                                                                     |
| GFRP                                                                              | DSS                                                                               | GFRP                                                                              | DSS                                                                               | GFRP                                                                              | DSS                                                                                | Side 1                                                                             | Side 2                                                                              | Side 1                                                                              | Side 2                                                                              |
|  |  |  |  |  |  |  |  |  |  |
| SL-AOJ-1                                                                          |                                                                                   | SL-AOJ-2                                                                          |                                                                                   | SL-AOJ-3                                                                          |                                                                                    | DL-AOJ-1                                                                           |                                                                                     | DL-AOJ-2                                                                            |                                                                                     |
| Only-bolted joints                                                                |                                                                                   |                                                                                   |                                                                                   |                                                                                   |                                                                                    |                                                                                    |                                                                                     |                                                                                     |                                                                                     |
| Brittle bearing + Shear-out                                                       |                                                                                   |                                                                                   |                                                                                   |                                                                                   |                                                                                    | Ductile bearing + Shear-out                                                        |                                                                                     |                                                                                     |                                                                                     |
| Side 1                                                                            | Side 2                                                                            | Side 1                                                                            | Side 2                                                                            | Lap side                                                                          | Back side                                                                          | Lap side                                                                           | Back side                                                                           | Lap side                                                                            | Back side                                                                           |
|  |  |  |  |  |   |   |   |  |  |
| DL-OBJ-1                                                                          |                                                                                   | DL-OBJ-2                                                                          |                                                                                   | SL-OBJ-1                                                                          |                                                                                    | SL-OBJ-2                                                                           |                                                                                     | SL-OBJ-3                                                                            |                                                                                     |

Fig. 7 Failure mode of single-joining-method joints.

| Hybrid bonded-bolted joints                                                         |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                       |                                                                                       |
|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| Cohesive + Shear-out                                                                |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     | Mixed + Ductile bearing                                                               |                                                                                       |
| Side 1                                                                              | Side 2                                                                              | Side 1                                                                              | Side 2                                                                              | Lap side                                                                            | Back side                                                                           | Lap side                                                                              | Back side                                                                             |
|  |  |  |  |  |  |  |  |
| DL-HBB-15-1                                                                         |                                                                                     | DL-HBB-15-2                                                                         |                                                                                     | SL-HBB-30-1                                                                         |                                                                                     | SL-HBB-30-4                                                                           |                                                                                       |
| Side 1                                                                              | Side 2                                                                              | Side 1                                                                              | Side 2                                                                              | Lap side                                                                            | Back side                                                                           | Lap side                                                                              | Back side                                                                             |
|  |  |  |  |  |  |  |  |
| DL-HBB-23-1                                                                         |                                                                                     | DL-HBB-23-2                                                                         |                                                                                     | SL-HBB-30-2                                                                         |                                                                                     | SL-HBB-30-5                                                                           |                                                                                       |
| Side 1                                                                              | Side 2                                                                              | Side 1                                                                              | Side 2                                                                              | Lap side                                                                            | Back side                                                                           |                                                                                       |                                                                                       |
|  |  |  |  |  |  |                                                                                       |                                                                                       |
| DL-HBB-30-1                                                                         |                                                                                     | DL-HBB-30-2                                                                         |                                                                                     | SL-HBB-30-3                                                                         |                                                                                     |                                                                                       |                                                                                       |

Fig. 8 Failure mode of hybrid bonded-bolted joints.

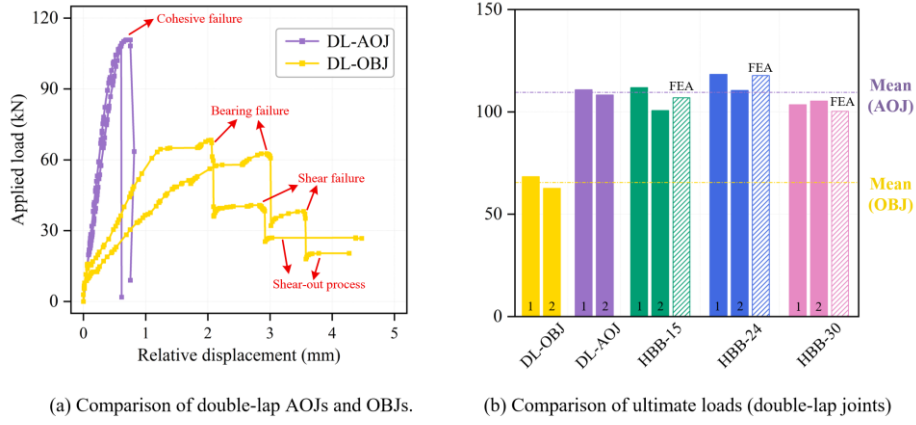


Fig. 9 Comparison of double-lap joints.

#### 4.1.2. End-distance effect

The numerical model captured the dominant failure sequence and final shear-out mode observed in the experiments (Fig. 10). As shown in Fig. 11, experimental data indicate the  $e_1/d_0 = 2.4$  configuration yielded the highest average load capacity. However, owing to limited sample sizes and data scattering in the  $e_1/d_0 = 1.5$  and  $3.0$  groups, this non-monotonic trend necessitated further numerical elucidation. Abaqus simulations accurately replicated the phenomena of higher and lower load capacities in the  $e_1/d_0 = 2.4$  and  $3.0$  groups, respectively. This corroborates that  $e_1/d_0$  critically influences localized stress distribution and damage evolution. Theoretically, while double-lap architectures mitigate eccentric bending and peel stresses relative to single-lap joints, stress concentrations (shear and peel) persist at overlap extremities, conforming to Volkersen's shear-lag theory, the Goland-Reissner model [24], and current numerical outputs. In these dissimilar GFRP and DSS joints, stiffness mismatches further exacerbate adhesive stress heterogeneity. Consequently, for a 88 mm overlap, altering the  $e_1/d_0$  modifies both the shear-out ligament length (toward the GFRP free edge) and the distance to the opposing high-stress adhesive zone. The end-distance effect is therefore not a singular shear-out area phenomenon, but a complex interplay between free-edge boundary effects and the coupling of hole-edge stress concentrations with adhesive high-stress zones. The ultimate capacity in these HBB joints is governed by characteristic distances that jointly modulate the coupling among the adhesive edge high-stress zone, the hole-edge contact pressure region, and the shear-out path. Mechanistically, maintaining the same  $d_0$ , smaller end distances restrict hole-edge stress diffusion, precipitating rapid shear-out upon contact pressure activation. Conversely, at larger end distances, the bolt hole's proximity to the adhesive high-stress zone exacerbates damage coupling, reducing peak loads. The intermediate  $e_1/d_0 = 2.4$  achieves an optimal equilibrium between these detrimental mechanisms, thereby yielding the maximum load-bearing capacity.

This variance is attributable to the bolt's effective stress zone, which is modulated by the  $e_1/d_0$  ratio. As shown in Fig. 12, pre-failure stress distributions extracted via numerical simulation reveal that adhesive failure initiates at the overlap edge of the GFRP laminate (distal to the joint end), stress magnitudes at this location across different  $e_1/d_0$  corroborate the experimental findings. Notably, the missing elements in the Fig. 12(a) contour plot are due to element deletion triggered by edge-initiated adhesive failure under an insufficient end distance.

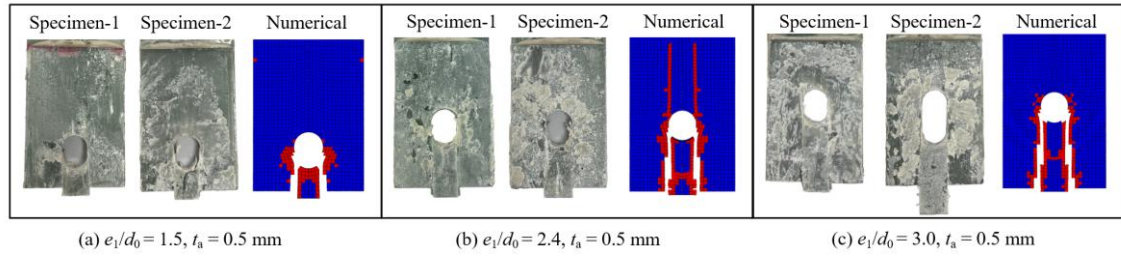


Fig. 10 Experimental and numerical failure modes of double-lap HBB joints.

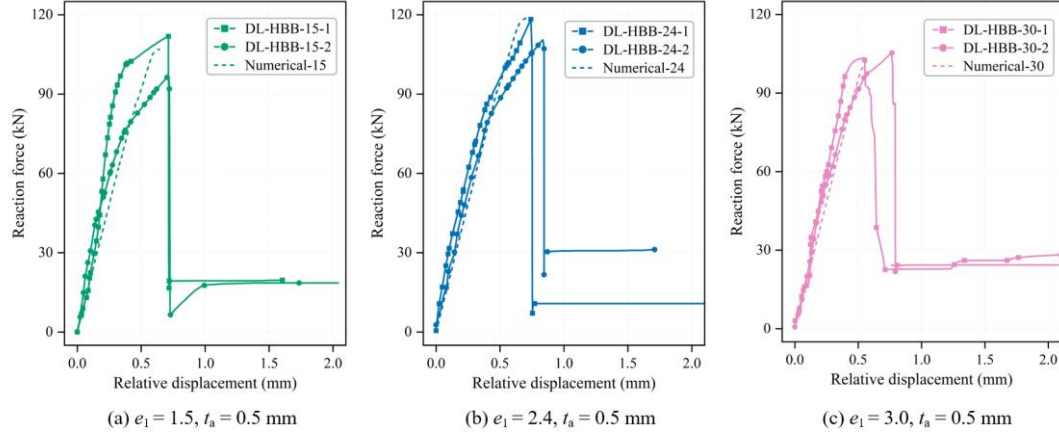


Fig. 11 Load-relative displacement curves of double-lap HBB joints.

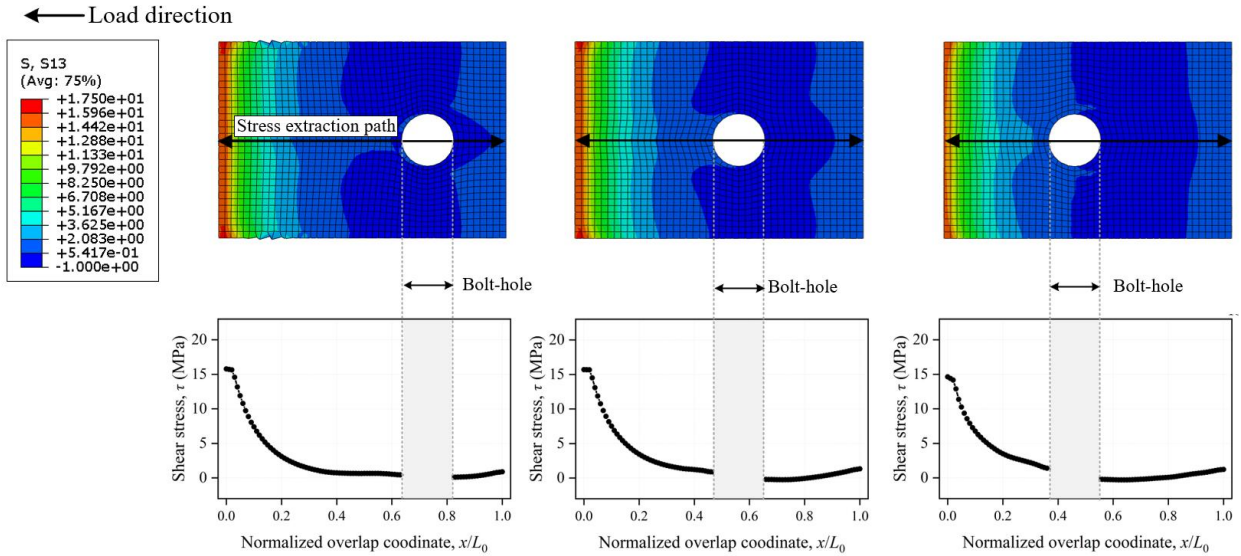


Fig. 12 Shear stress distribution in cohesive element at  $0.4 F_u$ .

#### 4.1.3. Double-bolt hybrid joints

For the double-lap double-bolt HBB joints, the horizontal arrangement produced higher peak loads than the vertical arrangement. This suggests that transversely arranged bolts act as parallel constraint points, enlarging the effective clamping/contact zone across the joint width and improving adhesive utilization before peak failure. By contrast, vertically arranged bolts act more like a series load-transfer path along the loading direction, where the stress fields around adjacent

holes interact with the longitudinal shear-out path. This sequential activation explains both the lower peak load and the additional small peak observed in the vertical specimens (Fig. 13).

In these hybrid joints, adhesive failure occurs at peak load, subsequently altering the bolt's interaction mechanism with the GFRP and ultimately culminating in shear-out failure. Crucially, in practical applications like marine engineering, adhesive failure inevitably permits seawater ingress. Consequently, this study defines the onset of adhesive failure as the point of global structural failure. Even if the bolt's load capacity is inferior to the adhesive's, it still acts as a "fail-safe" baseline after the adhesive fails, preventing the instantaneous disintegration of the structure. In practical engineering (especially in environments with vibration or fatigue loads), this is a crucial line of defense.

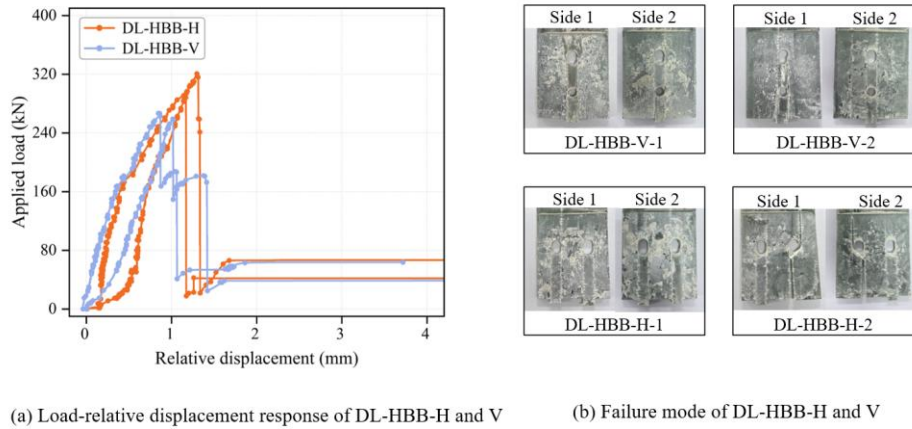


Fig. 13 Load-relative displacement curves of double-lap HBB joints with two bolts.

## 4.2. Single-lap joints

### 4.2.1. Failure modes and mechanism

Single-lap adhesive-only joints exhibited GFRP adherend failure, whereas only-bolted configurations failed via a combined bearing/shear-out mechanism. As shown in Fig.14, although localized bearing at the bolt-hole interface is universally evident, specimen SL-OBJ-1 displayed a notably truncated bearing plateau and minimal bearing area; this discrepancy is likely attributable to the inherent mechanical property dispersion of decommissioned composite blades. The divergence in ultimate failure modes between unidirectional (UD) and multidirectional (MD) single-lap joints indicates that layup configuration profoundly influences localized hole-edge damage progression in GFRP, a phenomenon detailed in the subsequent section.

Fig. 15 compares the normalized ultimate loads of single-lap and double-lap HBB joints with the same end distance. If the capacity scaled only with bonded area, the single-lap joint would reach approximately 0.5 of the double-lap capacity. However, single-lap joints are also affected by eccentric load transfer, opening gap and secondary bending, which would normally reduce the capacity below this ideal area-proportional value. The UD single-lap specimens reached a normalized load of approximately 0.53, suggesting that these adverse effects were partly offset by the limited gap opening and blind-bolt restraint. In contrast, the MD specimens dropped to approximately 0.42, indicating that the  $+45^\circ/0^\circ/-45^\circ$  layup induced larger gap opening and stronger bending. Therefore, the efficiency loss of single-lap HBB joints depends not only on the reduced bonded area, but also on the layup-dependent opening response.

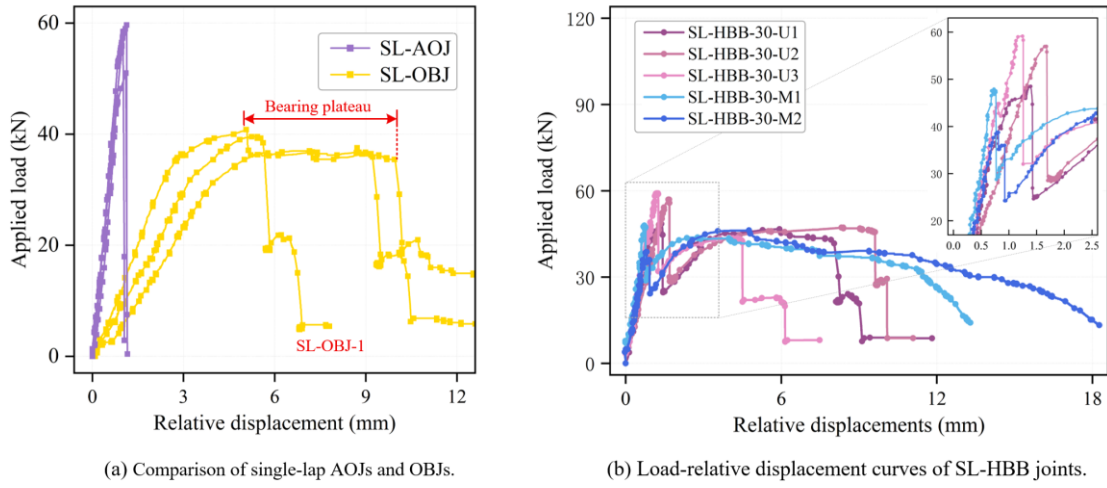


Fig. 14 Load-relative displacement curves of single-lap AOJs and OBJs.

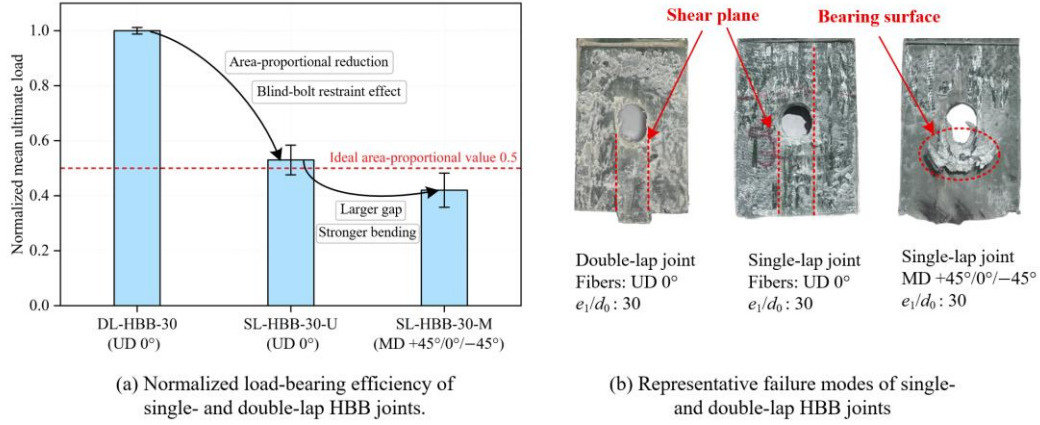


Fig. 15 Comparison of load-bearing efficiency and failure modes between single- and double-lap HBB joints.

#### 4.2.2. Effect of GFRP layup

For single-lap hybrid joints, adhesive layer failure consistently manifested as a mixed cohesive/adherend mode. Furthermore, magnified views (Fig. 14) reveal that at the onset of adhesive failure, the two specimens ultimately undergoing bearing failure exhibited lower ultimate loads. Unidirectional (UD) 0° GFRP single-lap hybrid joints demonstrated high stiffness, minimal deformation, and rapid fiber-aligned shear-out. Correlating with Fig. 16, multidirectional (+45°/0°/-45°) configurations exhibited greater localized deformation, extensive hole-edge bearing, and wider interfacial gaps, yet yielded lower peak capacities. Mechanistically, because fibers in pure 0° UD layups parallel the loading axis, hole-edge bearing damage rapidly propagates along the fibers, triggering brittle shear-out. Conversely,  $\pm 45^\circ/0^\circ/-45^\circ$  layups disperse stress and suppress unidirectional crack propagation, shifting the failure mechanism from shear-out to progressive hole-edge bearing; however, this reduced proportion of effective 0° fibers along the loading axis consequentially diminishes overall axial stiffness and ultimate capacity.

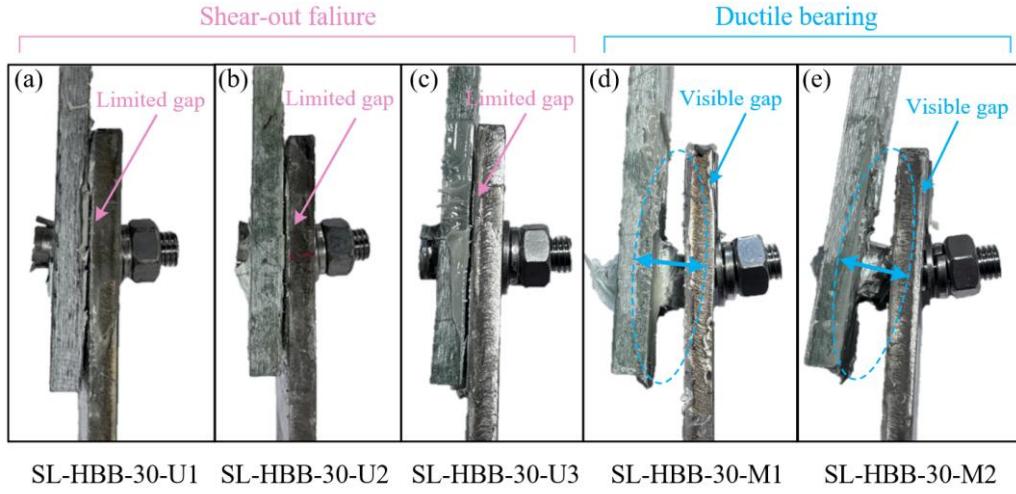


Fig. 16 Side views of single-lap joint failure modes for HBB joints.

### 4.3. Comparison with design codes

This section benchmarks the experimental load capacities of the repurposed GFRP-to-DSS hybrid joints against established fiber-reinforced polymer (FRP) design standards. Because current codes primarily target newly manufactured, quality-controlled FRPs, their direct application to decommissioned blades necessitates accounting for material dispersion, service-induced aging, and layup uncertainties. Consequently, the normative equations are utilized herein as a comparative design framework.

#### 4.3.1. Applicability and design philosophy of selected codes

The experimental results were evaluated against ASCE/SEI 74-23 [25], CEN/TS 19101: 2022 [26] and T/CECS 692-2020 [27]. ASCE/SEI 74-23 adopts a conservative predictive baseline for hybrid joints by applying pure bolted joint formulations [28], relegating the adhesive's contribution solely to initial stiffness enhancement. Conversely, CEN/TS 19101:2022 strictly prohibits the simple superposition of adhesive and bolt capacities due to inherent disparities in stiffness, slip initiation, and damage sequence. It mandates experimental or numerical validation, explicitly designating bolts as a "back-up system" to maintain a fail-safe structural state. The Chinese code T/CECS 692-2020 for pultruded composites permits evaluating hybrid joints as purely bolted connections while concurrently providing shear-loaded adhesive design formulations. Empirically, the initial peak load of the HBB joints in this study is predominantly governed by the adhesive layer, whereas post-peak residual capacity and ultimate failure modes depend on the GFRP hole-edge shear-out resistance. Consequently, a two-stage evaluation methodology is implemented herein: the initial peak is elucidated via the adhesive failure path, followed by analyzing the post-failure residual capacity and ultimate failure modes through the FRP shear-out resistance.

#### 4.3.2 Code-based resistance prediction and comparison with tests

To quantitatively examine the staged resistance mechanism identified in the tests, the code-based prediction was conducted at two levels. The first level evaluates the adhesive-controlled peak resistance of the HBB joints, while the second level assesses the FRP-side in-plane bolted resistance after adhesive degradation. Accordingly, the first peak resistance was compared with the predicted adhesive resistance, whereas the post-adhesive resistance was evaluated using the governing FRP-side failure modes, including shear-out, bearing and net-tension failure.

$$R_{peak-adhesive} = R_{adh} \quad (1)$$

$$R_{post-adhesive} = \min(R_{sh}, R_{br}, R_{nt}) \quad (2)$$

Where  $R_{peak-adhesive}$  is the predicted first peak resistance of HBB;  $R_{adh}$  represents the adhesive layer path resistance;  $R_{post-adhesive}$  denotes the in-plane bolted path resistance of FRP after adhesive layer failure;  $R_{sh}$  stands for the shear-out resistance;  $R_{br}$  is the bearing resistance; and  $R_{nt}$  signifies the net-tension resistance.

For consistency, the resistance equations from different codes were rewritten using a unified notation. Here,  $R_{adh}$  represents the adhesive-path resistance, and  $R_{sh}$ ,  $R_{br}$ , and  $R_{nt}$  denote the shear-out, bearing and net-tension resistances of the FRP bolted path, respectively. The calculated resistances were normalized by the experimental first peak load  $P_{peak,test}$  to clarify whether each resistance component corresponded to the adhesive-controlled peak stage or the post-adhesive mechanical redundancy stage.

$$\text{Shear-out resistance: } R_{sh} = C_{sh} n_c \left( e_1 - \frac{d_0}{2} + \alpha p_1 \right) t f_v \quad (3)$$

Where  $\alpha = 0$  for a single bolt row and  $\alpha = 1$  for two bolt rows arranged along the loading direction. For the horizontal double-bolt configuration,  $n_c = 2$  and  $\alpha = 0$ ; for the vertical double-bolt configuration,  $n_c = 1$  and  $\alpha = 1$ .

Table 4 Code-dependent coefficients for shear-out resistance

| Standard           | Single row $C_{sh,1}$ | Double row $C_{sh,2}$ |
|--------------------|-----------------------|-----------------------|
| ASCE/SEI 74-23     | 1.4                   | 1.4                   |
| CEN/TS 19101: 2022 | 1.5                   | 0.9                   |
| T/CECS 692-2020    | 1.4                   | 1.4                   |

$$\text{Bearing resistance: } R_{br} = C_{br} n_b d t f_{br} \quad (4)$$

Table 5 Code-dependent coefficients for bearing resistance

| Standard           | $C_{br}$     | Note                                |
|--------------------|--------------|-------------------------------------|
| ASCE/SEI 74-23     | $\zeta$      | Depends on thread/bearing condition |
| CEN/TS 19101: 2022 | $1.5/k_{cc}$ | Includes clearance/contact factor   |
| T/CECS 692-2020    | 1.0          | -                                   |

$$\text{Net-tension resistance: } R_{nt} = \frac{1}{k_{nt}} (w - n_h d_0) t f_t \quad (5)$$

$$\text{Shear adhesive resistance: } R_{adh} = 0.9 f_{adh} \cdot A_n \quad (6)$$

Where  $e_1$  is the end distance,  $d_0$  is the hole diameter,  $d$  is the bolt diameter,  $t$  is the GFRP thickness,  $w$  is the plate width,  $n_b$  is the number of bolts,  $n_h$  is the number of holes crossing the net section,  $n_c$  is the number of bolts in the transverse direction, and  $p_1$  is the bolt pitch in the loading direction.  $f_v$ ,  $f_{br}$ , and  $f_t$  denote the in-plane shear strength, bearing strength, and tensile strength of GFRP, respectively.  $C_{sh}$ ,  $C_{br}$ , and  $k_{nt}$  are code-dependent coefficients.

Table 6 summarizes the experimental first peak loads and code-predicted resistances of the HBB specimens. To avoid mixing different physical stages, all predicted resistances were normalized by the experimental first peak load. The ratio ( $R_{adh} / F_{peak}$ ) was used to evaluate the contribution of the adhesive path to the first peak load, whereas ( $R_{sh} / F_{peak}$ ) was used to assess the residual shear-out capacity of the GFRP ligament after adhesive degradation. For the single-bolt double-lap HBB specimens, ( $R_{adh} / F_{peak}$ ) was close to unity, indicating that the first experimental peak was mainly governed by the adhesive load path. In contrast, the values of ( $R_{sh} / F_{peak}$ ) predicted by the three codes were all much lower than unity, suggesting that the GFRP shear-out ligament could not sustain the adhesive-controlled peak load after adhesive failure. Therefore, the specimens experienced a rapid post-peak load drop and finally failed by shear-out, without developing a stable bearing plateau. With increasing end distance, ( $R_{sh} / F_{peak}$ ) increased gradually, indicating improved post-adhesive shear-out redundancy. Nevertheless, the predicted shear-out resistance remained lower than the experimental first peak load. This confirms that the peak resistance and final failure of HBB joints cannot be interpreted by a single bolted-connection resistance model, but should be understood through a staged mechanism: an adhesive-controlled peak followed by bolt-path redundancy.

For the double-bolt and single-lap specimens, the normalized results further indicate that joint configuration and laminate layup affect both adhesive utilization and shear-out redundancy. In particular, for single-lap specimens, the code-based single-lap reduction and in-plane strength parameters can only partially account for configuration effects, but cannot explicitly capture layup-dependent opening gap, secondary bending and eccentricity sensitivity. Therefore, the code equations are more suitable for identifying the potential bolt-hole failure mode after adhesive degradation than for directly predicting the first peak load of HBB joints.

Table 6 Normalized code-predicted adhesive and shear-out resistances of HBB joints

| Specimen    | Experimental<br>peak load | Predicted adhesive-<br>controlled peak | Predicted bolt-controlled residual redundancy |                                   |                                   |
|-------------|---------------------------|----------------------------------------|-----------------------------------------------|-----------------------------------|-----------------------------------|
|             |                           | T/CECS 692-2020                        | ASCE/SEI 74-23                                | CEN/TS 19101: 2022                | T/CECS 692-2020                   |
|             |                           | $F_{\text{peak}}$                      | $R_{\text{adh}} / F_{\text{peak}}$            | $R_{\text{sh}} / F_{\text{peak}}$ | $R_{\text{sh}} / F_{\text{peak}}$ |
| DL-HBB-15   | 106.25                    | 0.91                                   | 0.15                                          | 0.16                              | 0.15                              |
| DL-HBB-24   | 114.38                    | 0.85                                   | 0.26                                          | 0.28                              | 0.26                              |
| DL-HBB-30   | 104.38                    | 0.93                                   | 0.38                                          | 0.41                              | 0.38                              |
| DL-HBB-55-H | 307.78                    | 1.01                                   | 0.30                                          | 0.20                              | 0.61                              |
| DL-HBB-30-V | 262.67                    | 1.18                                   | 0.61                                          | 0.65                              | 0.61                              |
| SL-HBB-30-U | 54.86                     | 0.88                                   | 0.44                                          | 0.78                              | 0.73                              |

#### 4.3.3 Implications for reclaimed wind-turbine-blade GFRP joints

The existing FRP codes correctly predict the final shear-out failure of HBB joints but significantly underestimate the adhesive-governed first peak load, capturing only the residual bolted capacity. Given the high variability of reclaimed GFRP regarding layup, thickness, and service damage, design calculations must use region-specific measured properties instead of standard material strengths. Consequently, current code equations are best utilized for failure-mode identification and relative capacity assessment rather than absolute peak prediction.

Based on the present results, the design of reclaimed blade GFRP–metal HBB joints should not be based on a simple summation of adhesive and bolted resistances. A more appropriate approach is to distinguish two stages: an adhesive-controlled peak resistance and a post-adhesive mechanical redundancy governed by the FRP shear-out resistance. This staged framework explains the experimental first peak, the rapid post-peak load drop and the final shear-out failure, and provides a clearer assessment route for reusing repurposed wind-turbine-blade GFRP in marine and civil structural joints.

## 5. Conclusion

This study experimentally and numerically investigated hybrid bonded–bolted joints between GFRP reclaimed from a decommissioned wind turbine blade and 2205 duplex stainless steel for potential marine structural reuse. Based on the results, the following conclusions can be drawn.

1. The HBB joints exhibited a staged load-transfer and failure mechanism rather than a simple superposition of adhesive and bolted resistances. The first peak load was primarily governed by adhesive-layer failure, while the bolted path mainly affected the local stress field before adhesive damage and provided mechanical redundancy after adhesive degradation. Therefore, the load capacity of repurposed blade GFRP–DSS HBB joints should be interpreted through an adhesive-controlled peak stage followed by a bolt-controlled post-adhesive stage.

2. For double-lap single-bolt HBB joints, the end distance affected the coupling between the adhesive high-stress zone, the bolt-hole contact region and the GFRP shear-out ligament. The load capacity did not increase monotonically with end distance. Instead, an intermediate  $e_1/d_0$  produced the favourable balance by reducing premature shear-out while avoiding excessive interaction between the bolt-hole stress field and the adhesive edge-damage zone.

3. The double-bolt arrangement significantly influenced the effective load-transfer mechanism. The horizontal double-bolt configuration achieved higher ultimate loads than the vertical configuration, indicating a more favourable distribution of the bolt-related stress field and a higher effective utilization of the adhesive layer. Although the bolts did not dominate the first peak load, they acted as a fail-safe path after adhesive failure and delayed complete joint separation.

4. Single-lap blind-bolted tests demonstrated layup-dependent failure mechanisms crucial for blade repurposing. Unidirectional 0° specimens failed by rapid shear-out, whereas multidirectional +45°/0°/−45° layups exhibited a bearing-dominated response with larger deformation. Although this bearing behavior reduced coupon peak loads due to eccentricity and reduced 0° fibers, retaining these multidirectional laminates at the blind-bolt interface promotes stable damage progression and mitigates premature shear-out. Therefore, structural cut-out strategies should position specific multidirectional subregions at the bearing surface, moving beyond optimizations based solely on global axial strength.

5. The comparison with ASCE/SEI 74-23, CEN/TS 19101:2022 and T/CECS 692-2020 showed that existing FRP connection codes can reasonably identify the final FRP-side in-plane shear-out mode after adhesive degradation, but cannot directly predict the adhesive-controlled peak resistance of HBB joints. The predicted shear-out resistances were much lower than the experimental first peak loads, whereas the adhesive resistance was close to the measured peak load. Therefore, a staged assessment framework that separates adhesive-controlled peak resistance from bolt-controlled residual redundancy provides a clearer basis for designing repurposed blade GFRP–metal joints in marine and civil structural applications.

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